AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 17 · STUDENT LESSON

Prediction Is Not Fact

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 15: keep predictions separate from evidence used to judge them.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Separate a recorded prediction from the observation used to check it. Try the recall before opening notes.

Your learning target

I can separate a recorded **prediction** from the **observation** used to check it.

I can compare matching cases using a stated **outcome rule** and preserve unresolved **evidence**.

I can report checked results without counting missing **observation**s as successes or failures.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A **prediction** is a fallible **output** supported by clues.



Your target

LEARNING TARGET On Monday a team predicts 'rain on Tuesday'. On Tuesday they count a damp playground as success, although sprinklers ran. Write an **observation** rule that could have been agreed before Tuesday.

Learner name:

Readiness for the connected explanation

A model predicts far, but the distance record is unreadable. Can this case be called correct? What **evidence** would let you check it?

This week's end goal

By the end, I can keep predicted and observed outcomes separate and audit a disagreement.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Prediction Is Not Fact case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A rain **prediction** uses weather clues, but the later **observation** of rain or no rain is the **fact** that evaluates it.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

A **prediction** states what a model **output**s. An **observation** records what was measured or happened. To check a **prediction**, keep both records and compare the same case under the same **outcome rule**.

Writing a **prediction** into the **observation** column would make the check circular.

Use a fictional ramp investigation. Before each trial, Ramp1 predicts far for a ramp height of at least 6 cm and near otherwise. Its allowed heights are whole centimetres from 1 through 10.

After a toy rolls, a learner measures its travel distance. The **outcome rule** is far for a distance of at least 4 floor units and near otherwise. Ramp height and travel distance are different quantities.

The values in this lesson are invented practice records, not a physical law about ramps. Ramp1 is a fixed paper model for learning how to check **prediction**s.

Each trial has an ID so that its height, **prediction** and later measurement stay together. Keep the model version and record the **prediction** before opening the **observation** card.

STUDENT LESSON · 3 OF 18

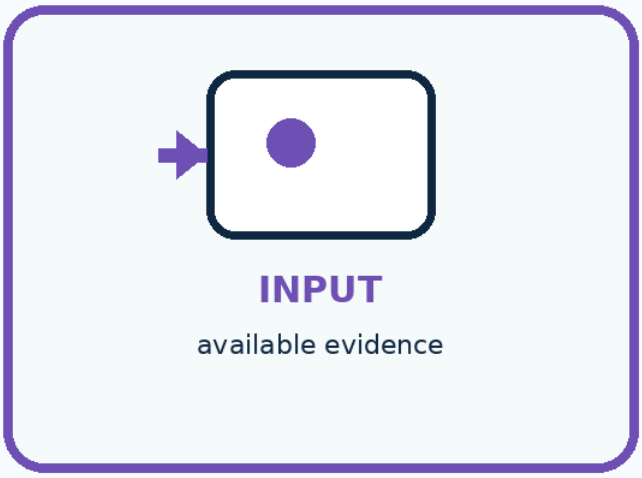
See how it works

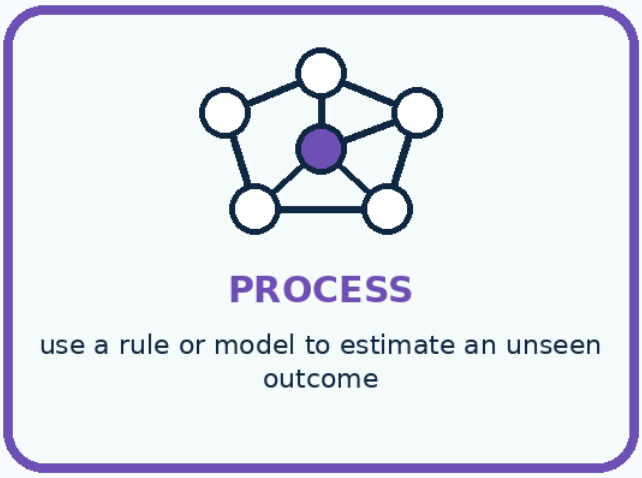
> Follow the mechanism for Prediction Is Not Fact. Connect each labelled input, deciding step and result.

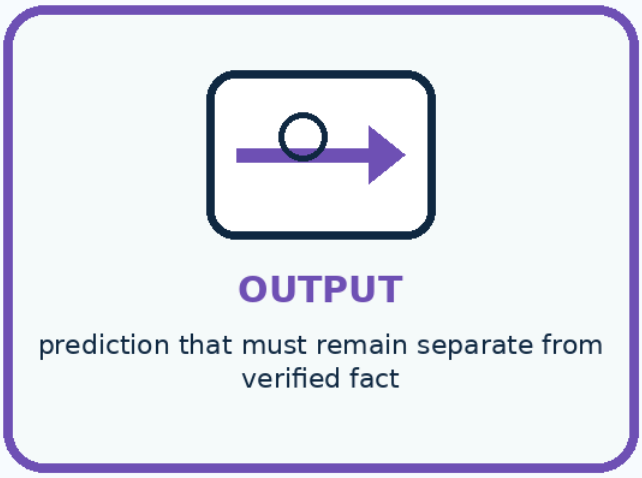
VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT available **evidence**

PROCESS use a rule or model to estimate an unseen outcome

OUTPUT prediction that must remain separate from verified **fact**

Key idea

CORE IDEA A **prediction** is a fallible **output** supported by clues.

How the mechanism works

UNDER THE HOOD **Prediction** estimates a target that is unknown at decision time. **Evidence** supports the estimate; **ground truth** is the later or independently established outcome.

Treating **prediction**s as labels can create self-confirming records.

Draw a fresh example with all three stages:

A closer explanation

Two records meet at a comparison

BEFORE THE TRIAL	AFTER THE TRIAL	COMPARISON
A: height 4 cm	Distance 3 units	Near matches near
B: height 6 cm	Distance 3 units	Far differs from near
C: height 8 cm	Distance 5 units	Far matches far

Ramp1 predicts from height: A near, B far, C far. The observation is travel distance. Apply the distance outcome rule separately, then compare categories for the same trial.

For B, the **input** 6 cm produces the **prediction** far. The later distance of 3 units produces the observed category near. This is a **mismatch**. Retain far as the original **prediction** and near as the observed category.

Changing the old **prediction** to near would hide the mistake rather than improve the original result.

For A and C, the categories match. These two matches provide **evidence** about these trials. They do not show that every future **prediction** will be correct.

Even a correctly computed **prediction** can disagree with an **observation** because following the model rule and matching the world are different checks.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Prediction Is Not Fact. Identify the detail that changes the conclusion.

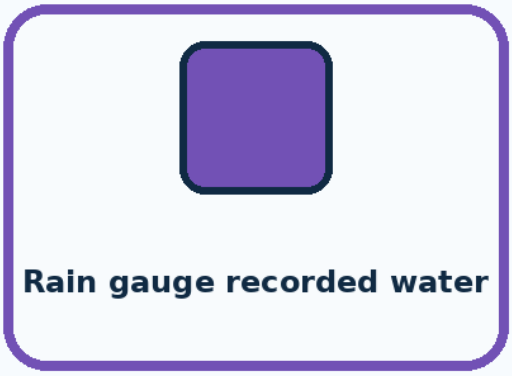
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Rain may start Clue: forecast **output**. Meaning: **prediction**.

The case to investigate

CASE 2 — Dark clouds Clue: observed condition. Meaning: **evidence**.

The case to investigate

CASE 3 — Rain gauge recorded water Clue: later measurement. Meaning: **fact** for evaluation.

The case to investigate

CASE 4 — Animal label from model Clue: unverified **output**. Meaning: **prediction**, not **fact**.

Keep this boundary clear

BOUNDARY Do not copy a model **prediction** into the answer column and later call it **ground truth**.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Keep score 4 but measure exactly 8 minutes.

Predicted result

Prediction: The observed label becomes quick and agrees with the **prediction**.

Why the result follows

Reason: Above 8 is strict: equality is quick under this policy.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Prediction Is Not Fact to explain a concrete decision, rather than only memorising definitions.

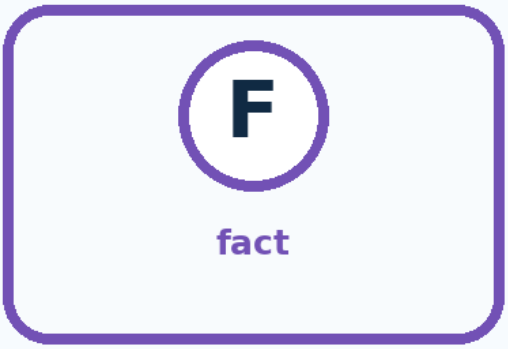
VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
prediction	estimate about an unknown outcome	
evidence	information that supports or challenges a claim	

Word	Meaning	My example
fact	claim established by direct or reliable checking	
ground truth	reference outcome used to evaluate a prediction	

Misconception detective

> **Find the unsupported claim**
>
> CLAIM TO REPAIR A confident **prediction** is a **fact**.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Observation	A recorded measurement or event used as evidence
Outcome rule	The agreed method for turning observed evidence into the category being checked
Mismatch	A prediction that differs from an established comparison outcome
Unassessed	A case whose prediction has not yet been judged against adequate outcome evidence

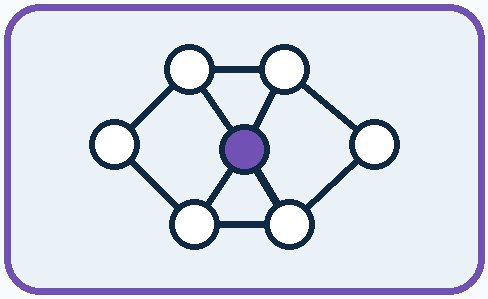
Follow a worked example

> Follow the worked trace for Prediction Is Not Fact, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE A rain **prediction** uses weather clues, but the later **observation** of rain or no rain is the **fact** that evaluates it.

Step 1

STEP 1 — OBSERVE available **evidence**

Step 2

STEP 2 — TRACE use a rule or model to estimate an unseen outcome

Step 3

STEP 3 — CONCLUDE **prediction** that must remain separate from verified **fact**

Model answer

MODEL REASONING Mark model or rule **output**s as **prediction**s, observable clues as **evidence**, and independently checked outcomes as **fact**s.

A **prediction** becomes evaluated against a **fact**; it does not become a **fact** merely because time passes.

A claim to avoid

DO NOT OVERCLAIM Do not copy a model **prediction** into the answer column and later call it **ground truth**.

What new **evidence** could change this answer?

A closer explanation

Record	Evidence available	Status under the outcome rule
D: height 5 cm; predicts near	Distance missing	Unassessed; request the trial record
E: height 7 cm; predicts far	Two records say 3 and 4 units	Unassessed; investigate the conflict
F: height 6 cm; predicts far	Verified distance exactly 4 units	Match; at least 4 includes 4

Worked explanation · Prediction Is Not Fact

For E, 3 units means near and 4 units means far. The disagreement can change the judgment. Check the trial ID, original measurement and recording procedure. Do not choose the distance that makes Ramp1 look better.

If the conflict cannot be resolved, retain both reports and mark the comparison **unassessed**.

Worked explanation · Prediction Is Not Fact

An **observation** is **evidence**, not an automatic guarantee of truth. A wrong trial ID, unclear unit or copying error can make the comparison unreliable.

If a supported correction is made, retain the original entry, the correction, its source and the reason. Remeasuring a new trial does not reveal what happened in the old one.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first **mismatch**. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Prediction Is Not Fact

A fictional drying trial uses a fixed rule and later measures what actually happened.

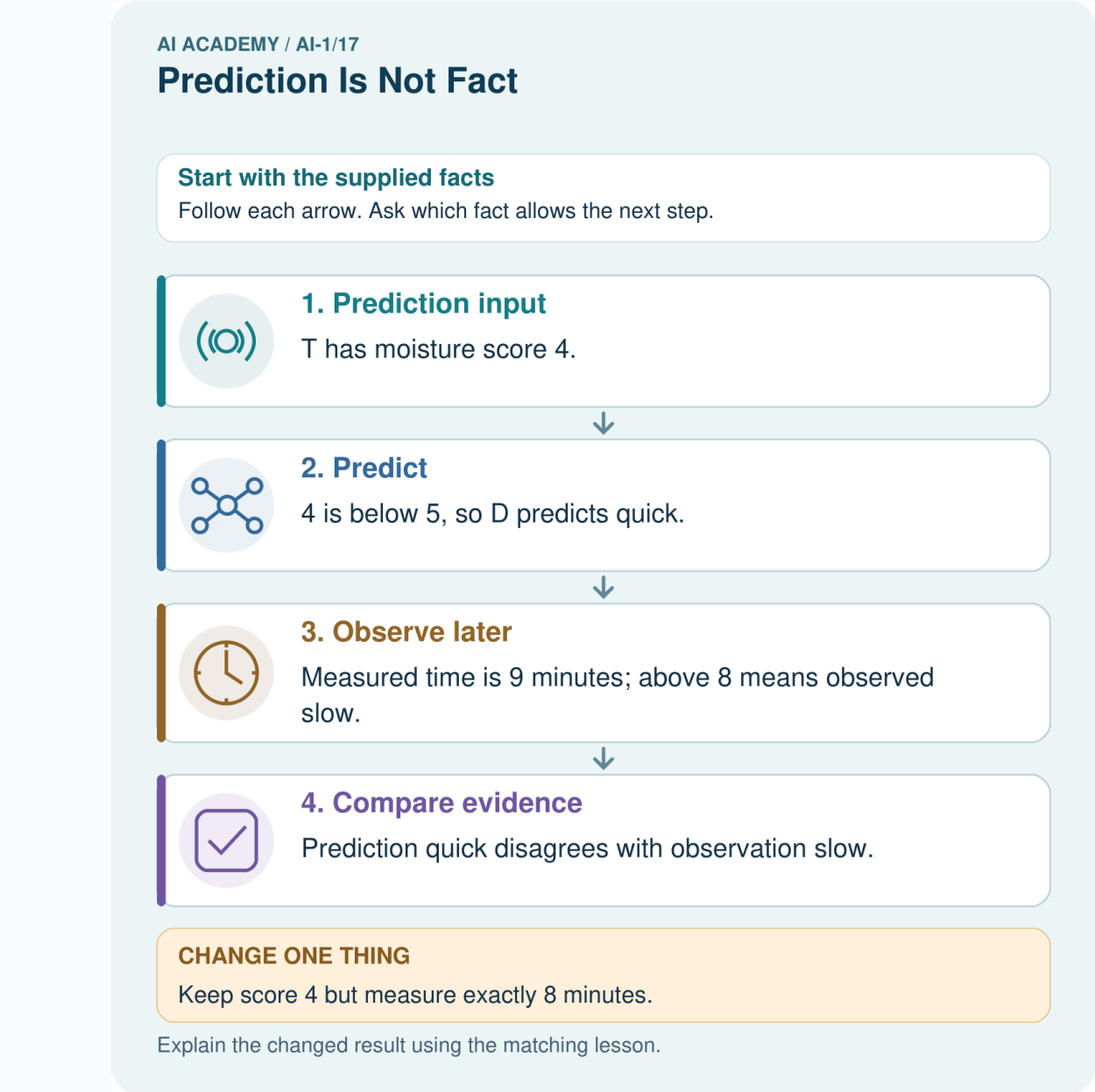
The facts we will use

Supplied facts

Model D predicts slow when moisture score is at least 5; quick otherwise.

Supplied facts

Observation policy: slow if drying takes above 8 minutes; quick otherwise. Trial T has score 4 and measured time 9 minutes.



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Prediction input	T has moisture score 4.
2. Predict	4 is below 5, so D predicts quick.
3. Observe later	Measured time is 9 minutes; above 8 means observed slow.
4. Compare evidence	Prediction quick disagrees with observation slow.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

The moisture score generates a **prediction**; it is not the observed drying time. The later measurement supplies separate **evidence** under the stated **observation** policy. Save both values and their units.

Altering the recorded time to match D would destroy the evaluation rather than repair the model.

What this example does not establish

Keep the stated limit

One failure does not prove every **prediction** is wrong; one agreement does not establish reliability on future trials.

Keep the stated limit

Read the labelled picture: Which field changes the **observation** label, and which field changes the model **prediction**?

Keep the stated limit

Change one thing: Keep score 4 but measure exactly 8 minutes. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Prediction Is Not Fact claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Rain may start	forecast output	prediction
Dark clouds	observed condition	evidence
Rain gauge recorded water	later measurement	fact for evaluation
Animal label from model	unverified output	prediction, not fact

Repair two tempting claims

Challenge claim A

CLAIM A A confident **prediction** is a **fact**.

Repair:

Challenge claim B

CLAIM B **Evidence** and **prediction** are two words for the same thing.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Start a **mismatch** investigation by checking the **input**, trial ID, **prediction** calculation and **observation** procedure. A **mismatch** alone does not identify which part caused the problem.

If all records are sound, this case is **evidence** that the model **prediction** did not match the outcome.

If you revise the model after seeing B, record a new version. A new **prediction** for B can help explain the change, but it is no longer a check on an unseen case.

Preserve Ramp1's result and use new reserved trials to evaluate the revised version, following Week 15's method.

> **A learner changes B's old prediction from far to near after reading its distance, then reports three matches out of three. Explain what is wrong with the report and how to preserve an honest record**

STUDENT LESSON · 8 OF 18

Practise together

> Practise Prediction Is Not Fact with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A **prediction** is a fallible **output** supported by clues.

Use a fresh example to explain **prediction**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: **Evidence** and **prediction** are two words for the same thing.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Consider only trials A through E. Three have established outcomes: A, B and C. Two of those three **prediction**s match and one does not. D and E are **unassessed**.

Report two matches out of three assessed trials, with two more trials **unassessed**. Do not report four successes by treating the unknowns as correct, or two out of five as if every trial had a known outcome.

Write a results record for A through E. Include each prediction, its observed category or unresolved evidence, its comparison status, and a summary that names both assessed and unassessed counts. Explain why D and E cannot be counted as correct

The missing cases might be different from the checked cases. The result describes the assessed group; it does not settle performance on all five or on future trials.

Collect the missing **evidence** where possible, document what remains unknown, and keep the original model version visible. Week 23 will develop performance fractions more fully.

Practise with less help: Check a boundary without changing the rule

Use workbook W2 for the connected example "**Prediction**s and **observation**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Prediction**s and **observation**s. Proposed explanation: "Running the rule correctly proves the **prediction** correct". Decide whether this explanation is supported.

Point to the step or supplied **fact** that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Which field changes the **observation** label, and which field changes the model **prediction**?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

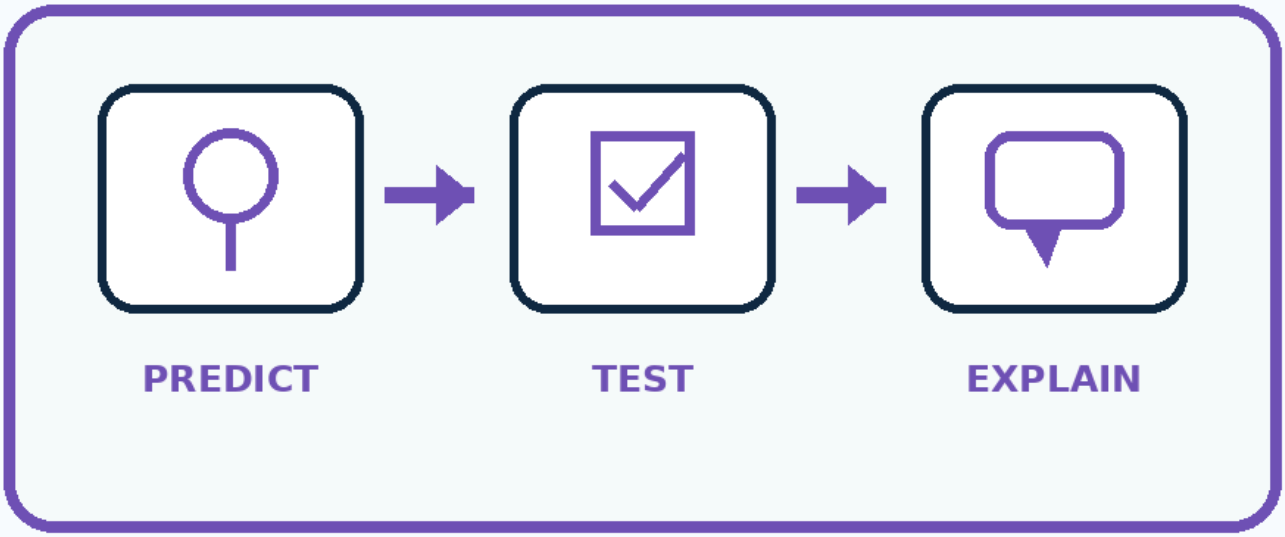
Run the investigation

> Investigate Prediction Is Not Fact. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Prediction-Evidence-Fact Sort

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Separate **prediction**, **evidence**, and **fact** in four scenarios.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

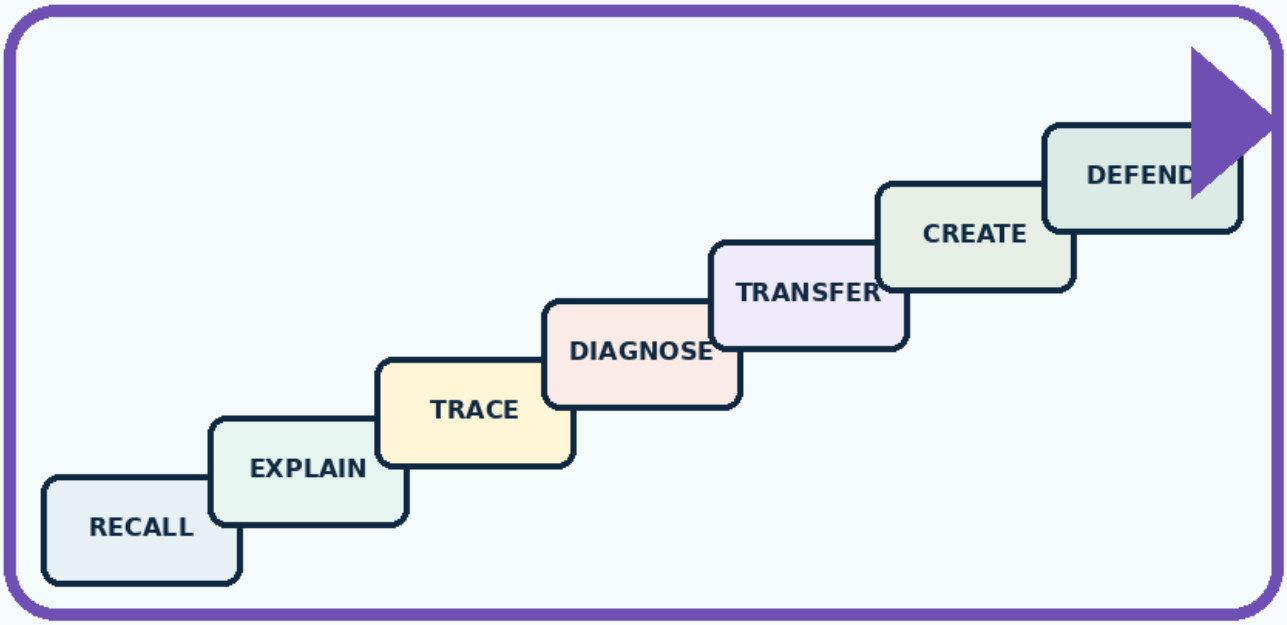
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Prediction Is Not Fact. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **prediction**.

EXPLAIN A **prediction** is a fallible **output** supported by clues.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: A confident **prediction** is a **fact**.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

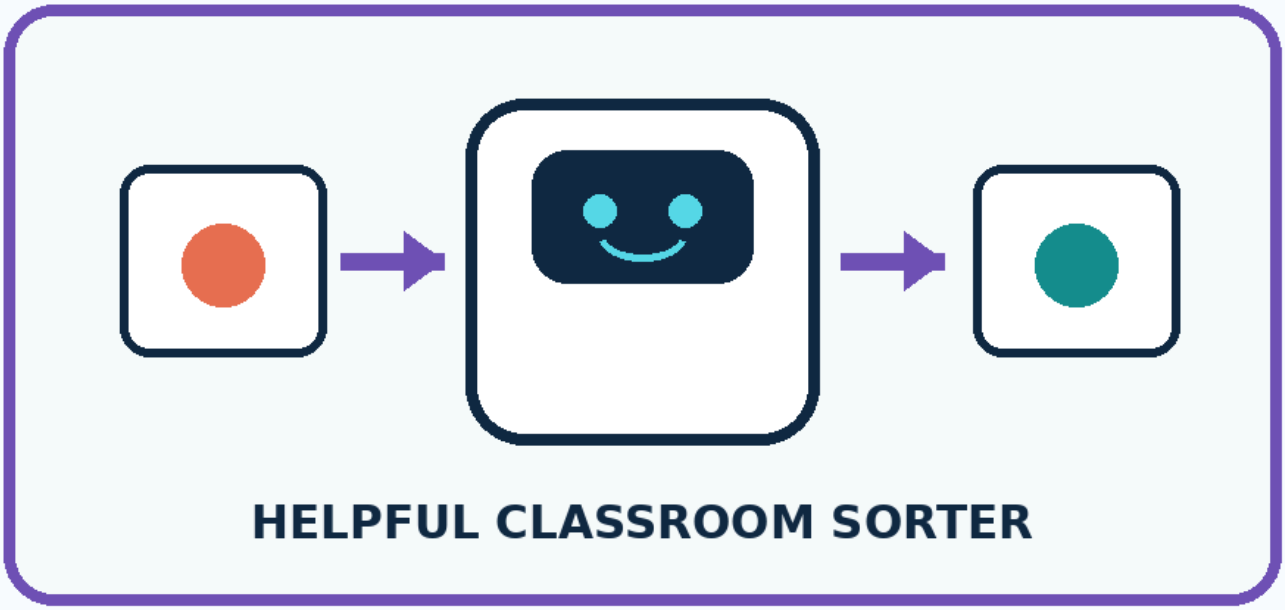
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Label the sorter's suggested bin as a **prediction** until a human confirms the final placement.

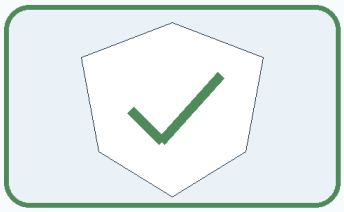
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN Do not copy a model **prediction** into the answer column and later call it **ground truth**.

My project decision:

Evidence for your project

In sorter trials, record predicted categories before collecting reference **evidence**, then retain disagreements for investigation.

STUDENT LESSON · 12 OF 18

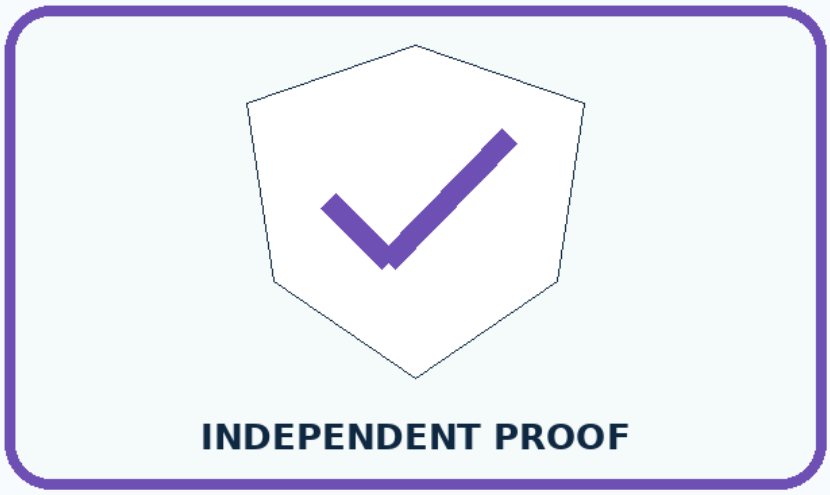
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Separate **prediction**, **evidence**, and **fact** in four scenarios and explain how one **prediction** could later be checked.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Mark model or rule **output**s as **prediction**s, observable clues as **evidence**, and independently checked outcomes as **fact**s.

A **prediction** becomes evaluated against a **fact**; it does not become a **fact** merely because time passes.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Confidence Is Not Correctness.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Prediction Is Not Fact case for supported practice. Repeated examples do not become fresh assessments.

Week 17 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A weather helper predicts tomorrow's rain.

The question

> Is this a **fact** about tomorrow? How can it be checked?

Worked explanation

It is a **prediction**. Record it now and compare with a predefined rain **observation** tomorrow. Useful **prediction**s are not guarantees.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

A prediction becomes checkable evidence

Evidence or stage	Value or result
Today	Record tomorrow's rain prediction
Tomorrow	Observe using an agreed rain rule
Compare	Prediction versus observation

A weather helper predicts tomorrow's rain.

> Is this a **fact** about tomorrow? How can it be checked?

It is a **prediction**. Record it now and compare with a predefined rain **observation** tomorrow. Useful **prediction**s are not guarantees.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Transfer the check to drying predictions

In a fictional paper test, Dry1 predicts slow for a moisture score of at least 7, quick otherwise. Allowed scores are whole numbers 1 through 10.

The observed outcome is slow if drying time is greater than 10 minutes, quick otherwise. Records are P: score 6, time 12; Q: score 7, time 10; R: score 9, time 14; S: score 5, time missing.

Record all **prediction**s, observed categories and comparison statuses. Summarise matches out of assessed cases, name the **unassessed** count, and limit the conclusion.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 17 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A model predicts rain tomorrow. Tomorrow is dry. Was the **prediction** an **observation** of tomorrow? Explain what new **evidence** settles the result.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Label the sorter's suggested bin as a **prediction** until a human confirms the final placement.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

On Monday a team predicts 'rain on Tuesday'. On Tuesday they count a damp playground as success, although sprinklers ran. Write an **observation** rule that could have been agreed before Tuesday.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Prediction Is Not Fact, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional drying trial uses a fixed rule and later measures what actually happened.

By the end, I can	How I will show it
Separate a recorded prediction from the observation used to check it	Explain the deciding step, show a trace or test, and state one limit.
Compare matching cases using a stated outcome rule and preserve unresolved evidence	Explain the deciding step, show a trace or test, and state one limit.
Report checked results without counting missing observations as successes or failures	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Prediction Is Not Fact without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Separate a recorded **prediction** from the **observation** used to check it

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional drying trial uses a fixed rule and later measures what actually happened.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Prediction** quick disagrees with **observation** slow.

1. Predict

2. Compare **evidence**

3. Observe later

4. **Prediction input**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare **evidence Prediction** quick disagrees with **observation** slow. The moisture score generates a **prediction**; it is not the observed drying time.

The later measurement supplies separate **evidence** under the stated **observation** policy. Save both values and their units. Altering the recorded time to match D would destroy the evaluation rather than repair the model.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Keep score 4 but measure exactly 8 minutes. Which conclusion is supported by the supplied **fact**s?

1. The observed label becomes quick and agrees with the **prediction**.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those **fact**s.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The observed label becomes quick and agrees with the **prediction**. Above 8 is strict: equality is quick under this policy.

Repair a missing building block

Review Boundaries and 'Not Sure' (ai-1/16). Goal: Identify a specific source of uncertainty and explain how it affects a **prediction**. Short practice: Which original block needs a better width before accepting or rejecting it?

What should the reviewer do?. Return to this week and explain what you changed.

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Trace 0 cm and explain why clarification is different from predicting short.. Return to this week and explain what you changed.

Review Repeating Efficiently (ai-1/11). Goal: Trace a loop by testing its condition before every repetition. Short practice: Trace a fresh start of 9. How many additions occur, and must the report equal 10?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

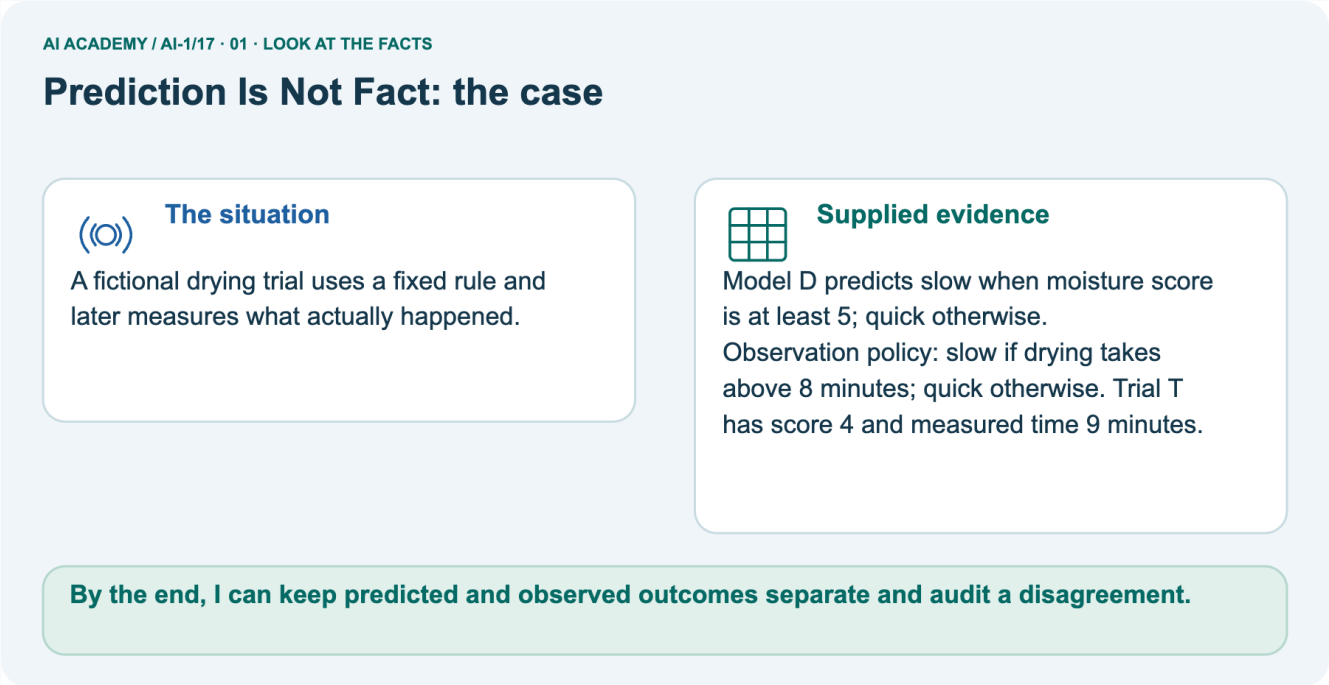
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Prediction Is Not Fact graphic using its labels, deciding evidence and relevant limitation.

01. Prediction Is Not Fact: the case



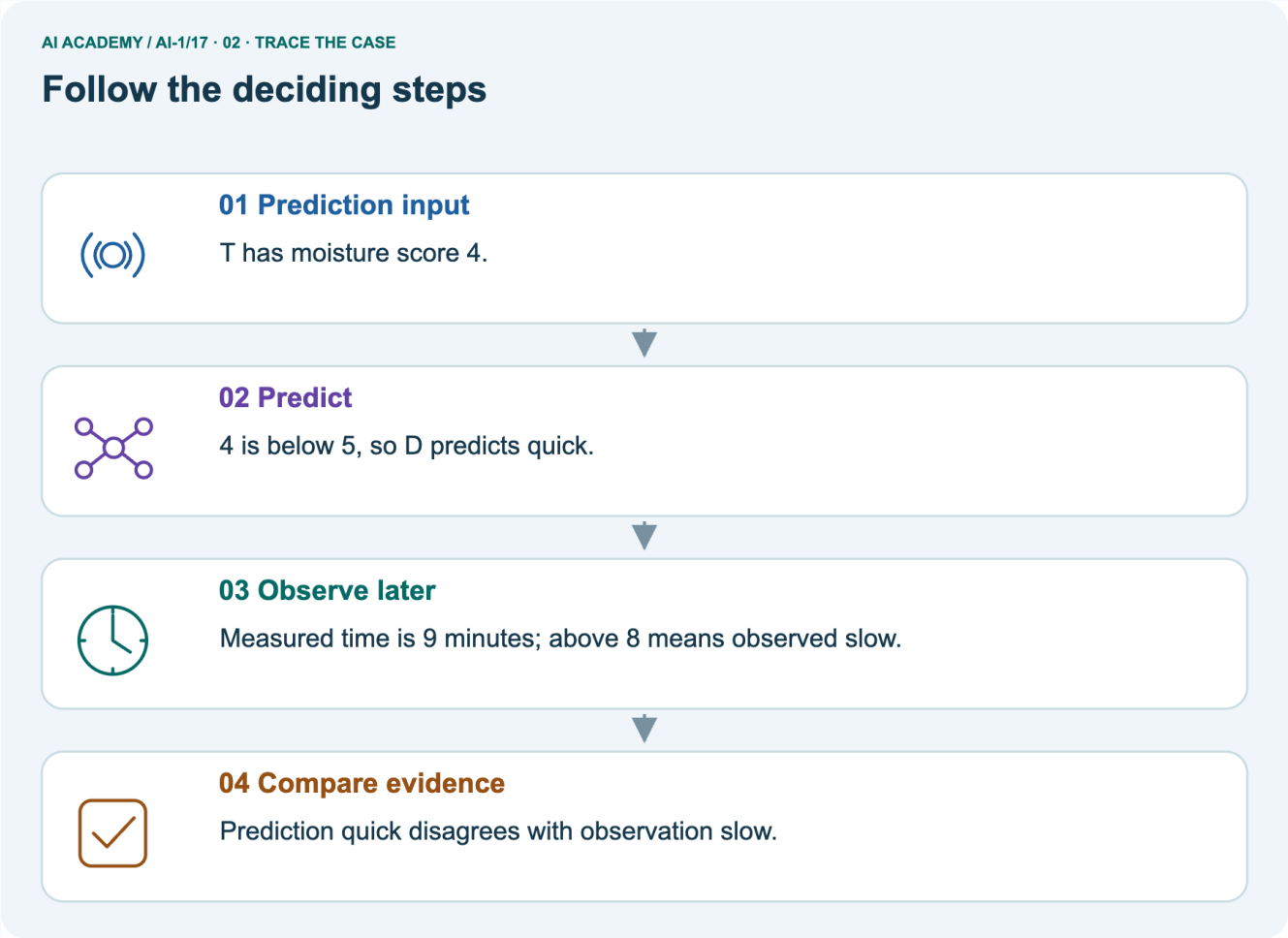
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional drying trial uses a fixed rule and later measures what actually happened. Supplied **fact**s: Model D predicts slow when moisture score is at least 5; quick otherwise.

Observation policy: slow if drying takes above 8 minutes; quick otherwise. Trial T has score 4 and measured time 9 minutes.

> Point to the given **fact** that matters most. Explain why it matters.

02. Follow the deciding steps

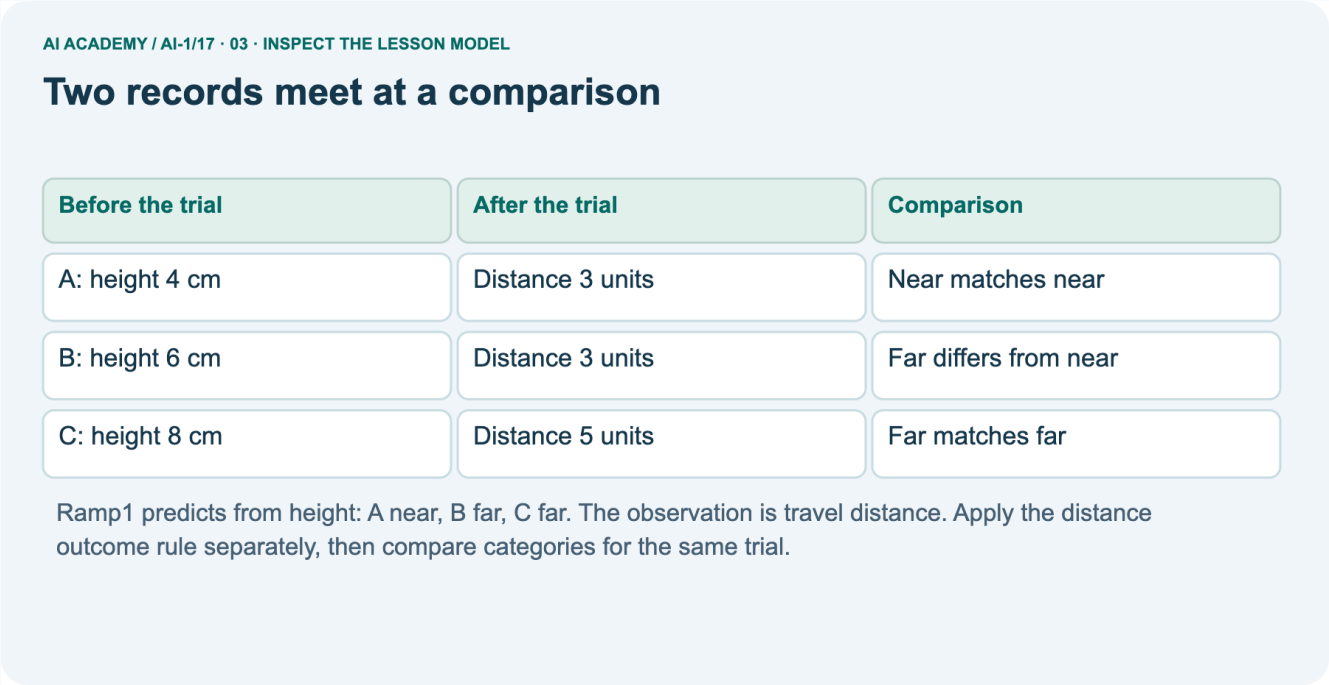


Follow the labelled picture with the matching explanation. Open the vector diagram

Prediction input: T has moisture score 4. → Predict: 4 is below 5, so D predicts quick. → Observe later: Measured time is 9 minutes; above 8 means observed slow. → Compare **evidence**: **Prediction** quick disagrees with **observation** slow.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Two records meet at a comparison

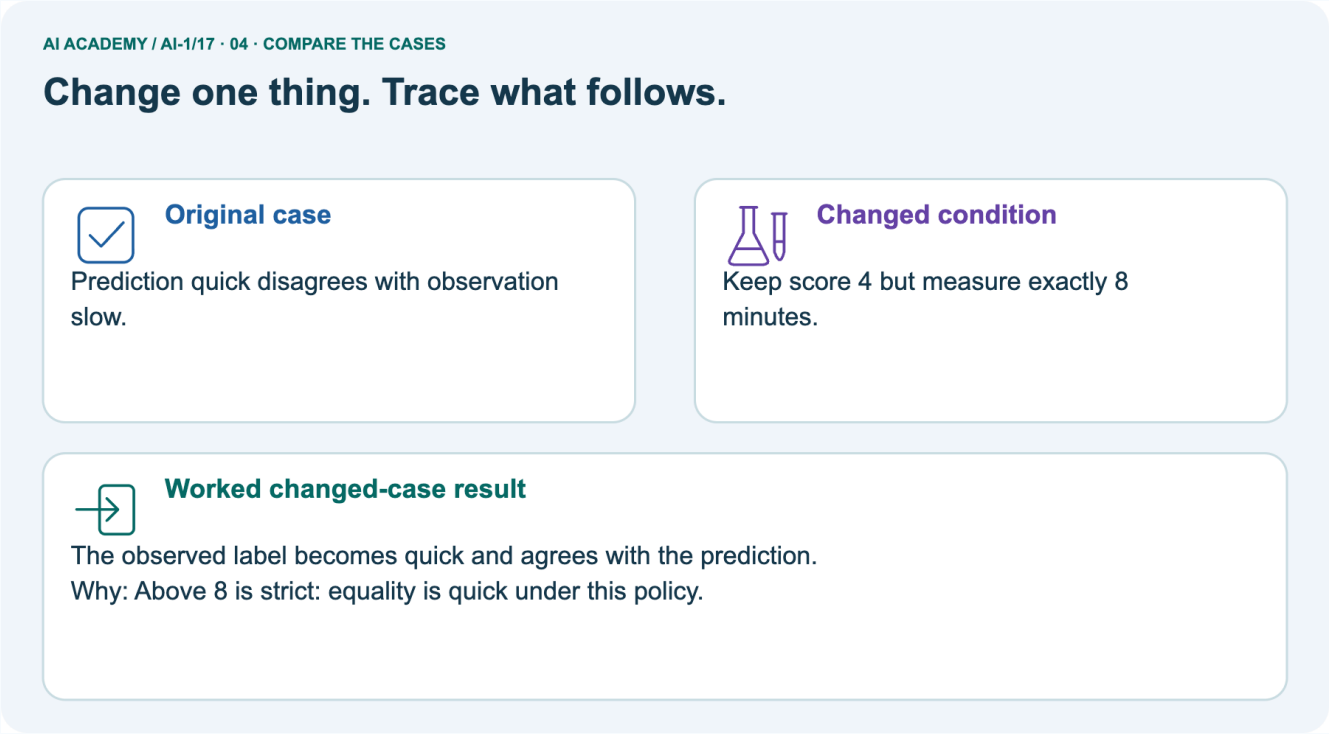


Follow the labelled picture with the matching explanation. Open the vector diagram

Two records meet at a comparison. A: height 4 cm | Distance 3 units | Near matches near. B: height 6 cm | Distance 3 units | Far differs from near. C: height 8 cm | Distance 5 units | Far matches far

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



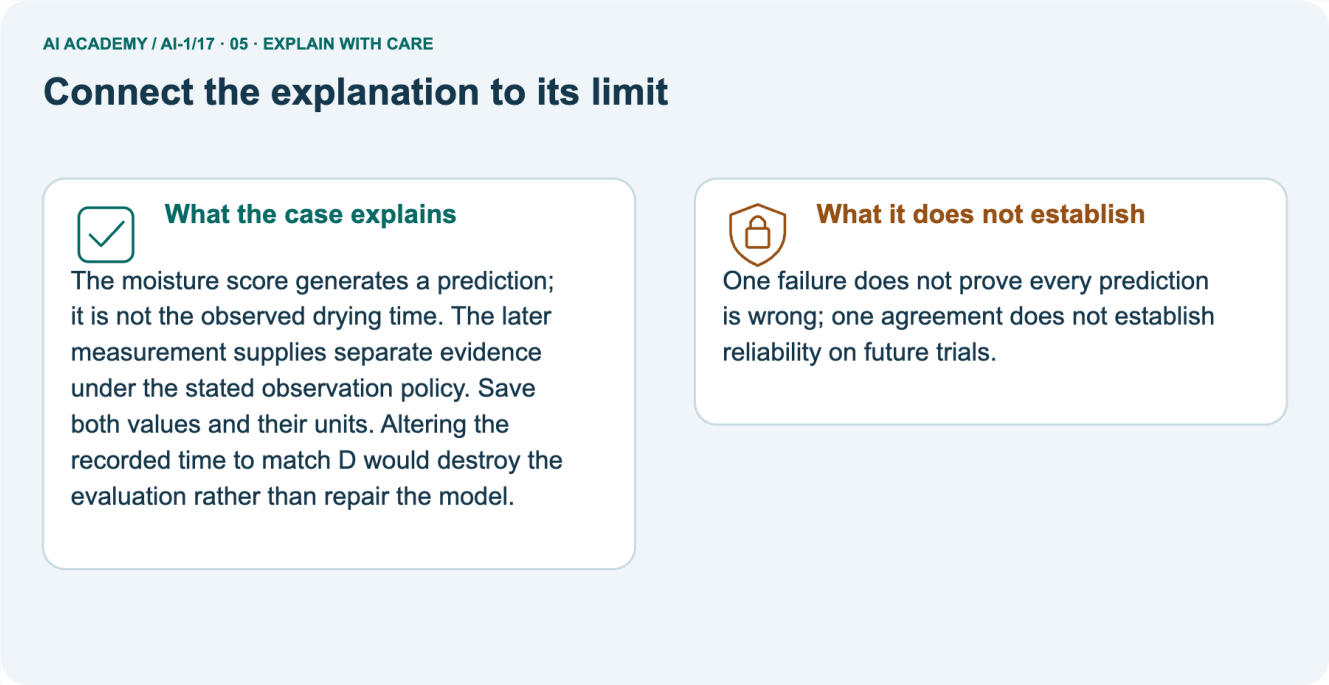
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Prediction** quick disagrees with **observation** slow. Changed condition: Keep score 4 but measure exactly 8 minutes. Expected result: The observed label becomes quick and agrees with the **prediction**.

Reason: Above 8 is strict: equality is quick under this policy.

> Predict before checking the worked changed-case result. Name what changed and what the supplied **fact**s allow you to conclude.

05. Connect the explanation to its limit



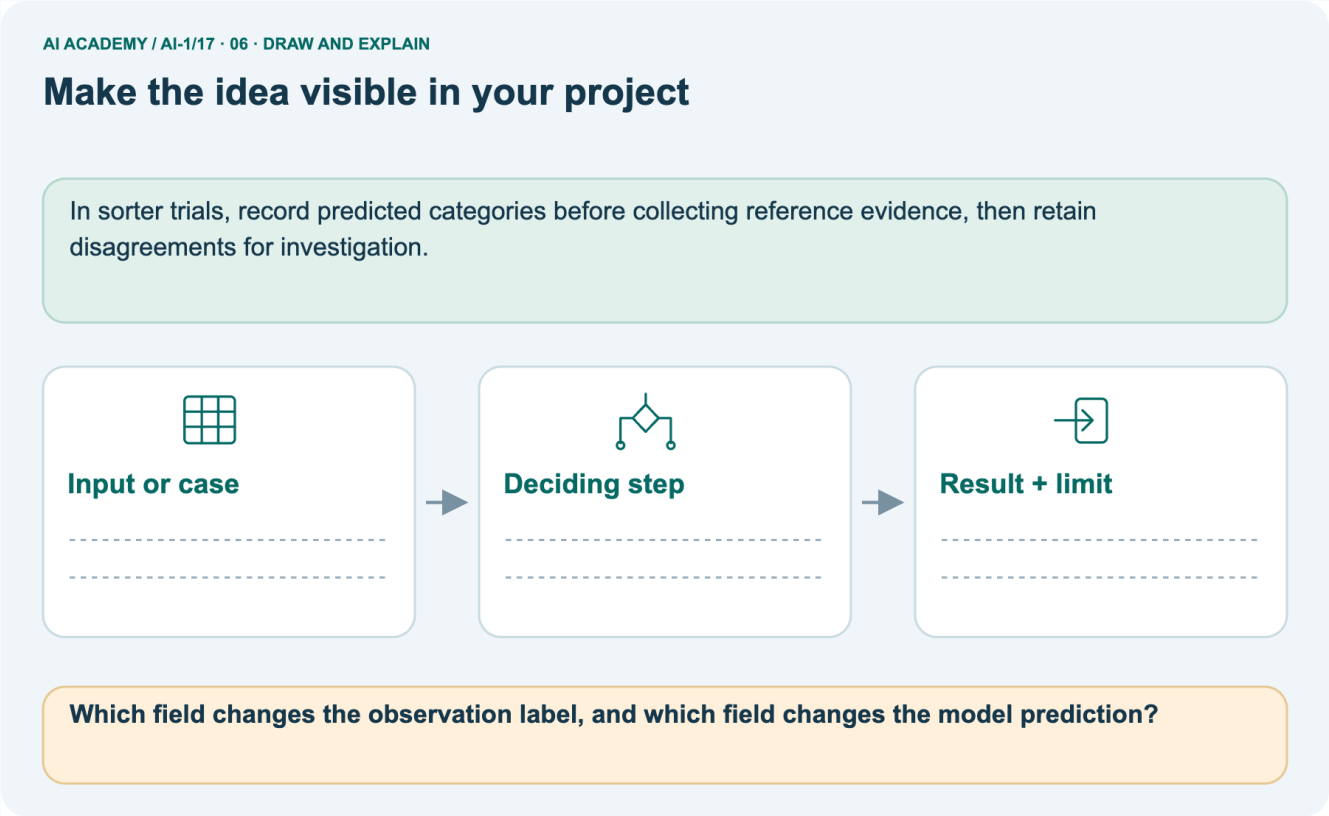
Follow the labelled picture with the matching explanation. Open the vector diagram

The moisture score generates a **prediction**; it is not the observed drying time. The later measurement supplies separate **evidence** under the stated **observation** policy. Save both values and their units.

Altering the recorded time to match D would destroy the evaluation rather than repair the model. Limit: One failure does not prove every **prediction** is wrong; one agreement does not establish reliability on future trials.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

In sorter trials, record predicted categories before collecting reference **evidence**, then retain disagreements for investigation. Draw your **input**, deciding step, result and limit.

Supported question: Which field changes the **observation** label, and which field changes the model **prediction**?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.